

SL-GPS Validation Report: Methyl Formate (CH_3OCHO) Mechanism Reduction via Supervised Learning — ThInK 1.0

SL-GPS Automated Validation Framework

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Abstract

This report presents a validation of the Supervised Learning – Global Pathway Selection (SL-GPS) method for adaptive chemical mechanism reduction applied to methyl formate (CH_3OCHO) combustion using the ThInK 1.0 mechanism (160 species, 1089 reactions). The complete pipeline is documented: training data generation via GPS, neural network training, and validation against the detailed mechanism across 8 test cases spanning temperatures of 800–1500 K, pressures of 1–20 atm, and equivalence ratios of 0.7–1.3. All training parameters, species classifications, and results are reported for full reproducibility.

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1 Introduction

The SL-GPS method trains neural networks on Global Pathway Selection (GPS) data to dynamically predict which chemical species are important during combustion simulations. At each timestep, the ANN evaluates the current thermochemical state and activates only the predicted species, producing an adaptive reduced mechanism that varies in size throughout the simulation.

This report validates SL-GPS on the ThInK 1.0 mechanism [1] for methyl formate (CH_3OCHO , also known as methyl methanoate), a simple oxygenated fuel of interest as a biodiesel surrogate and a key intermediate in dimethyl ether (DME) and dimethoxymethane (DMM) oxidation.

1.1 Mechanism Overview

Table 1: ThInK 1.0 mechanism summary.

Property	Value
Name	ThInK 1.0
Species	160
Reactions	1089
Target fuels	$\text{C}_1\text{--C}_3$ oxygenated fuels
Format	Cantera YAML

2 Methodology

2.1 SL-GPS Pipeline

The pipeline consists of three stages:

1. **Data generation:** Run N autoignition simulations with randomized initial conditions (T , P , ϕ). At each timestep interval, apply GPS ($\alpha = 0.001$) to label which species are important.
2. **ANN training:** Train an ensemble of neural networks on the GPS-labeled data. Select the best model by validation loss.
3. **Validation:** Run new autoignition cases with the detailed mechanism and simultaneously with SL-GPS (ANN-guided reduction). Compare temperature, HRR, species profiles, and ignition delay.

2.2 Species Classification

Before training, species are partitioned into three groups based on occurrence frequency in the GPS training data:

- **Always** ($> 99\%$ occurrence): unconditionally included in all reduced mechanisms
- **Variable** ($1\text{--}99\%$): predicted by the ANN at each timestep
- **Never** ($< 1\%$ occurrence): permanently excluded

Only the *variable* species are outputs of the neural network.

2.3 Neural Network Architecture

- **Input:** temperature, pressure, and 14 selected species mole fractions (normalized via MinMaxScaler)
- **Hidden layers:** 2 fully-connected layers $\times$ 32 neurons with ReLU activation
- **Output:** sigmoid activation (one output per variable species)
- **Loss:** binary cross-entropy; **Optimizer:** Adam
- **Training:** ensemble of 8 models in parallel, best selected by validation loss
- **Early stopping:** patience = 30 epochs, max = 200 epochs

3 CH₃OCHO — ThInK 1.0

3.1 Training Configuration

Table 2: CH₃OCHO training configuration summary.

Parameter	Value
<i>Mechanism</i>	
Mechanism	ThInK 1.0
Species / Reactions	160 / 1089
<i>Data Generation</i>	
Training cases	15 (parallel execution)
Temperature range	800–1600 K (uniform)
Pressure range	1–20 atm (log-uniform)
Equivalence ratio ϕ	0.7–1.4 (uniform)
GPS α	0.001
Total training samples	707
<i>Species Classification</i>	
Always species	14 (unconditionally included)
Variable species (ANN outputs)	63
Never species	83 (permanently excluded)
<i>ANN Training</i>	
Input features	T, P + 14 species mole fractions
Architecture	2 hidden layers $\times$ 32 neurons
Ensemble size	8 parallel trainings
Max epochs / Early stopping	200 / patience 30

Input species (ANN features): CH₃OCHO, O₂, H₂O, CO₂, CO, OH, H, H₂, CH₄, CH₂O, CH₃OH, HO₂, CH₃O, HCO

3.2 Validation Cases

Table 3: CH₃OCHO validation conditions (4 baseline + 4 extended).

Case	T_0 (K)	P (atm)	ϕ	t_{end} (s)
T1000_P1	1000	1	1.0	0.1
T1200_P1	1200	1	1.0	0.01
T1200_P10	1200	10	1.0	0.005
T1500_P1	1500	1	1.0	0.002
<i>Extended cases (difficult conditions)</i>				
T800_P20	800	20	1.0	0.5
T900_P10	900	10	1.0	0.5
T1100_P1_rich	1100	1	1.3	0.05
T1400_P5_lean	1400	5	0.7	0.01

3.3 Results — Baseline Cases

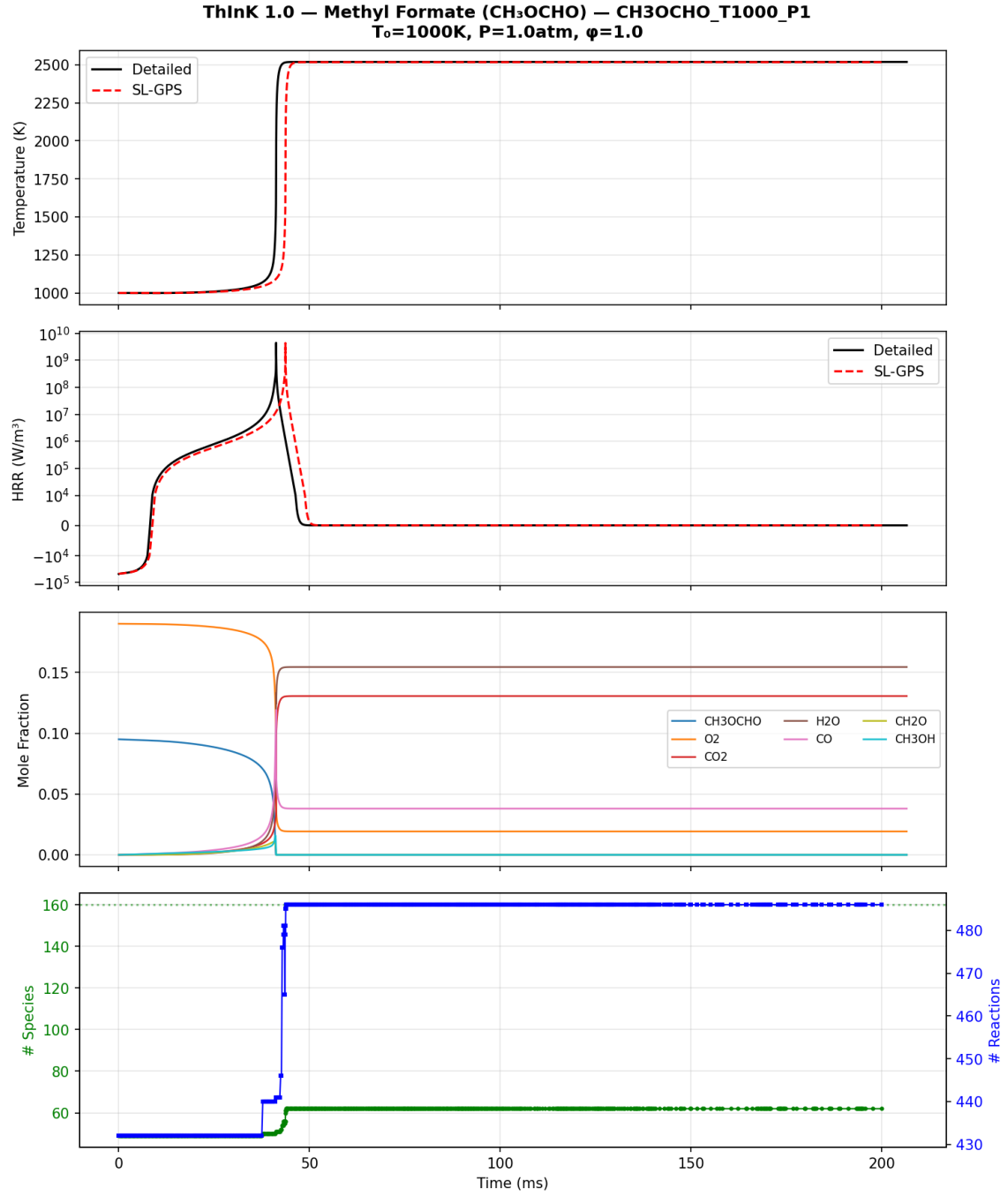


Figure 1: CH_3OCHO Case 1: $T_0 = 1000\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.0$. Ignition delay error: 6.0%.

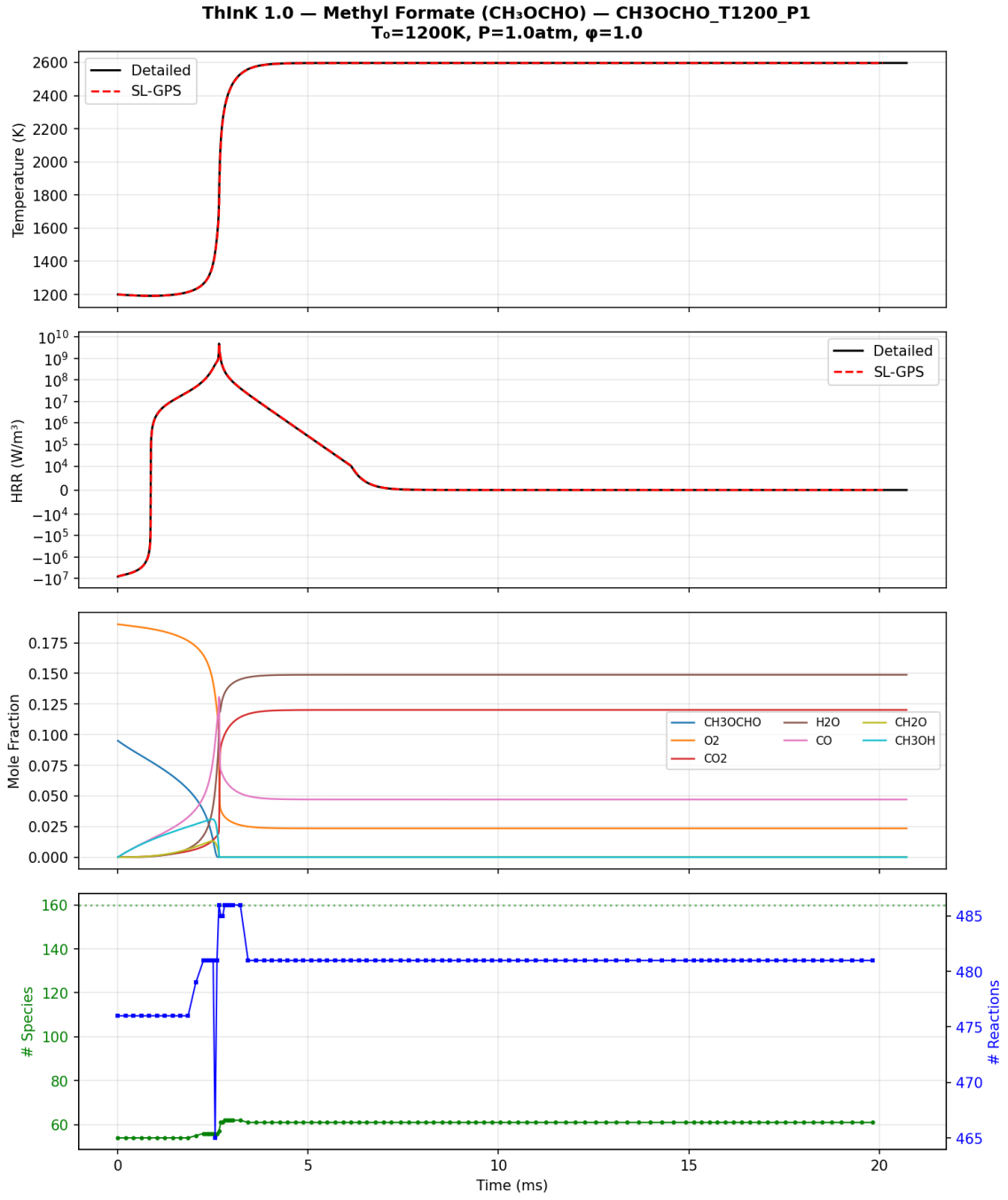


Figure 2: CH₃OCHO Case 2: $T_0 = 1200$ K, $P = 1$ atm, $\phi = 1.0$. Ignition delay error: 0.2%.

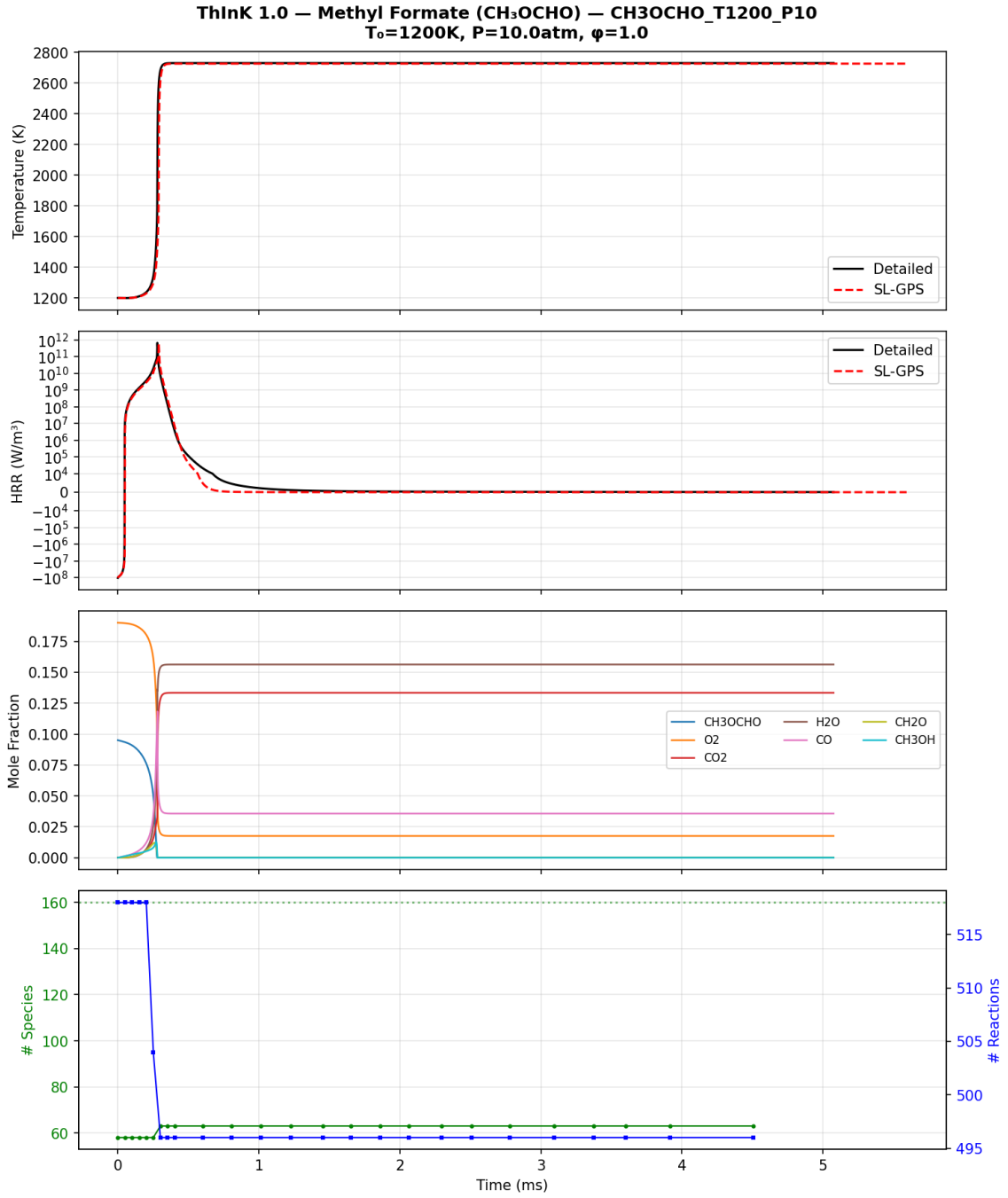


Figure 3: CH₃OCHO Case 3: $T_0 = 1200$ K, $P = 10$ atm, $\phi = 1.0$. Ignition delay error: 3.4%.

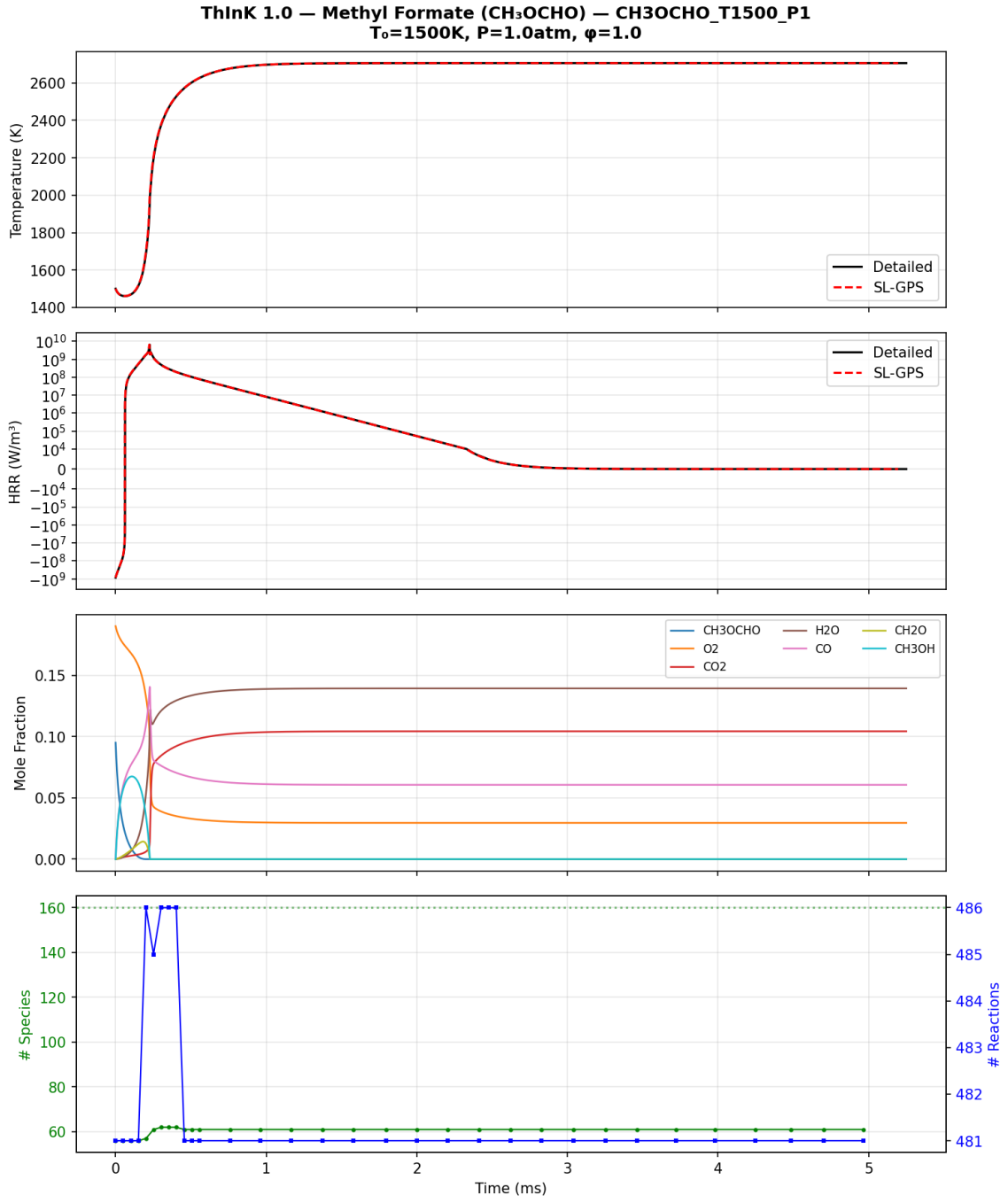


Figure 4: CH₃OCHO Case 4: $T_0 = 1500\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.0$. Ignition delay error: 0.9%.

3.4 Results — Extended Cases

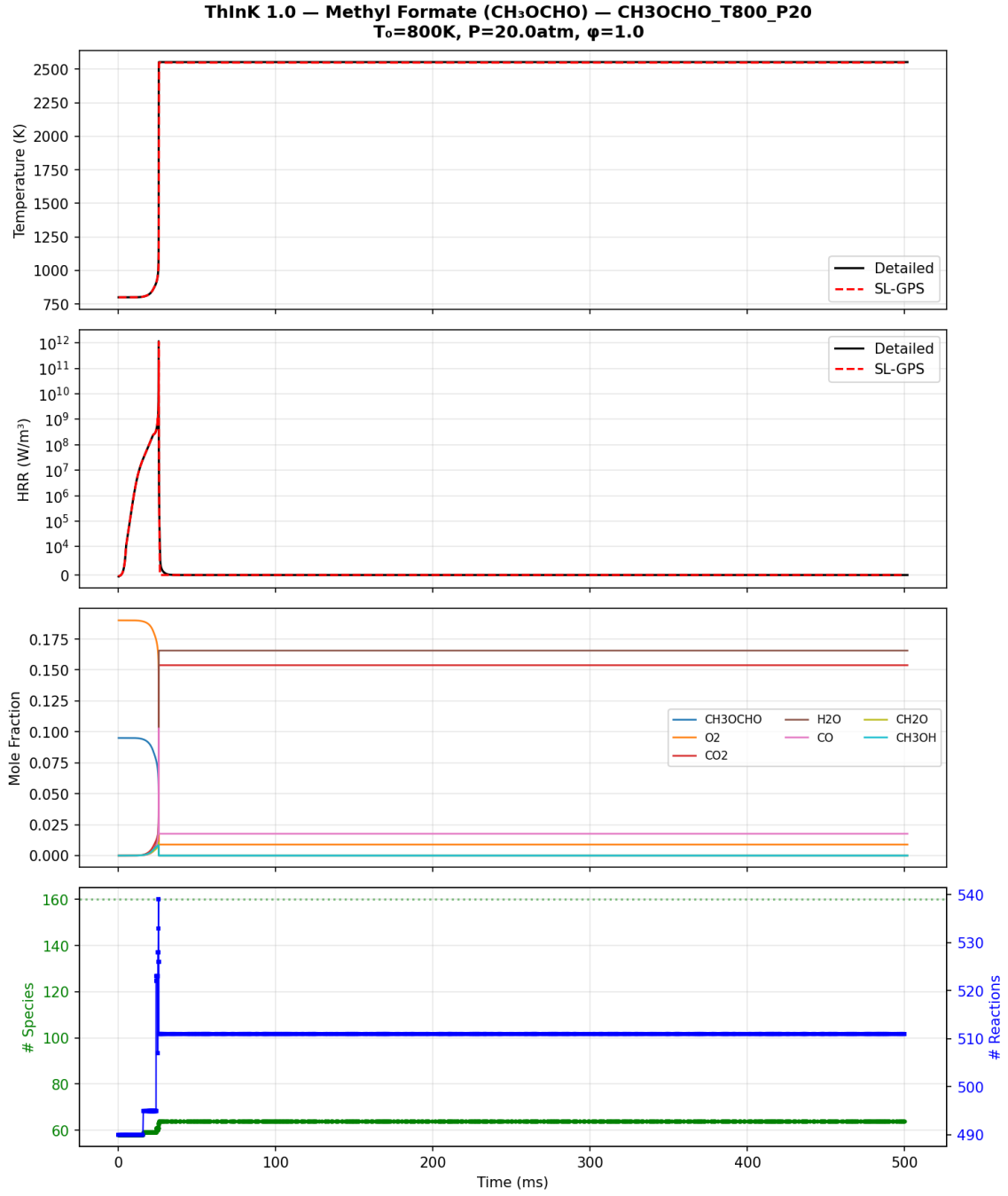


Figure 5: CH₃OCHO Case 5: $T_0 = 800\text{K}$, $P = 20\text{atm}$, $\phi = 1.0$. Ignition delay error: 0.3%.

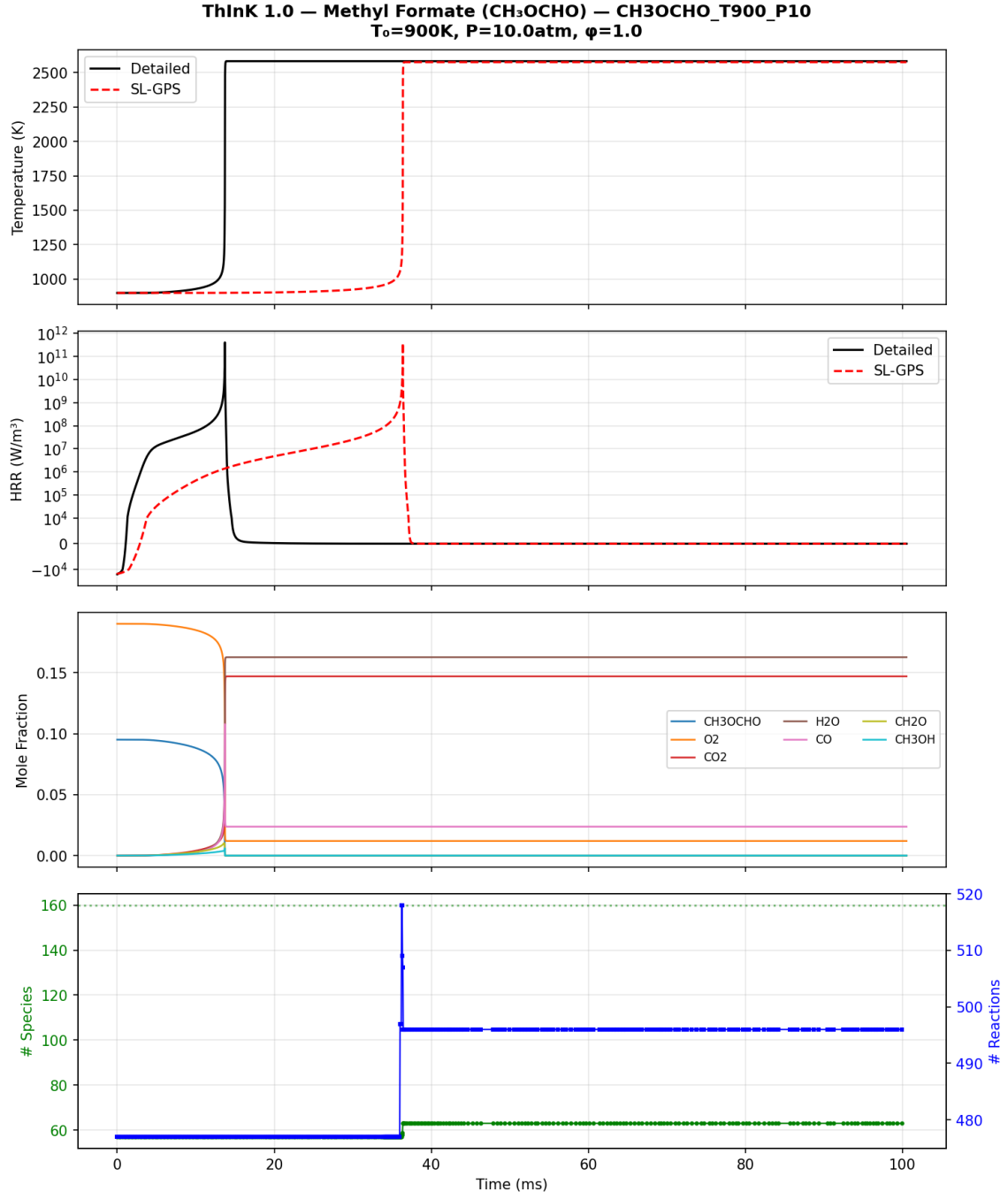


Figure 6: CH₃OCHO Case 6: $T_0 = 900$ K, $P = 10$ atm, $\phi = 1.0$. Ignition delay error: 165.2%. **Failed** — low-temperature NTC regime insufficiently covered by training data.

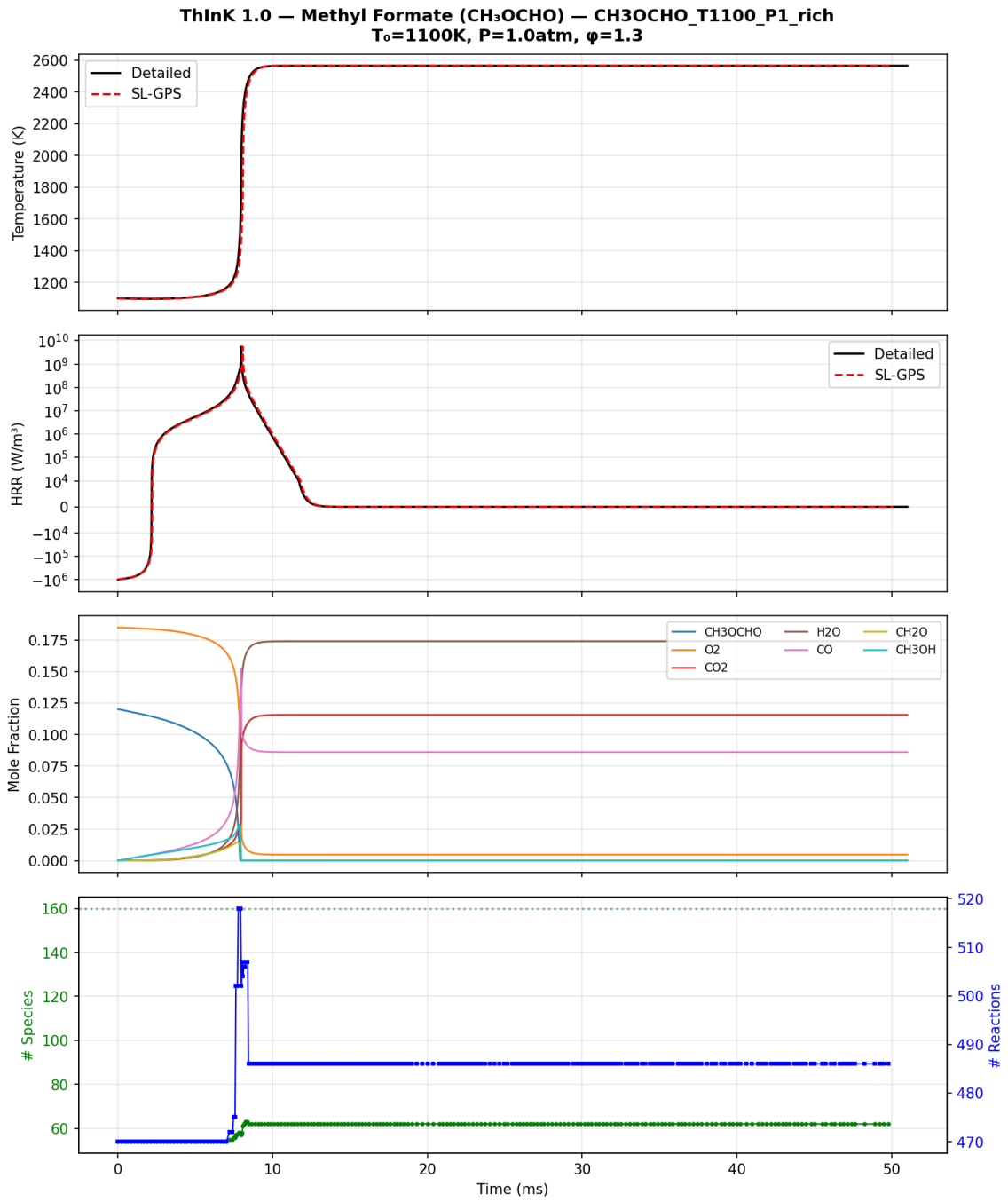


Figure 7: CH₃OCHO Case 7: $T_0 = 1100\text{ K}$, $P = 1\text{ atm}$, $\phi = 1.3$. Fuel-rich. Ignition delay error: 1.3%.

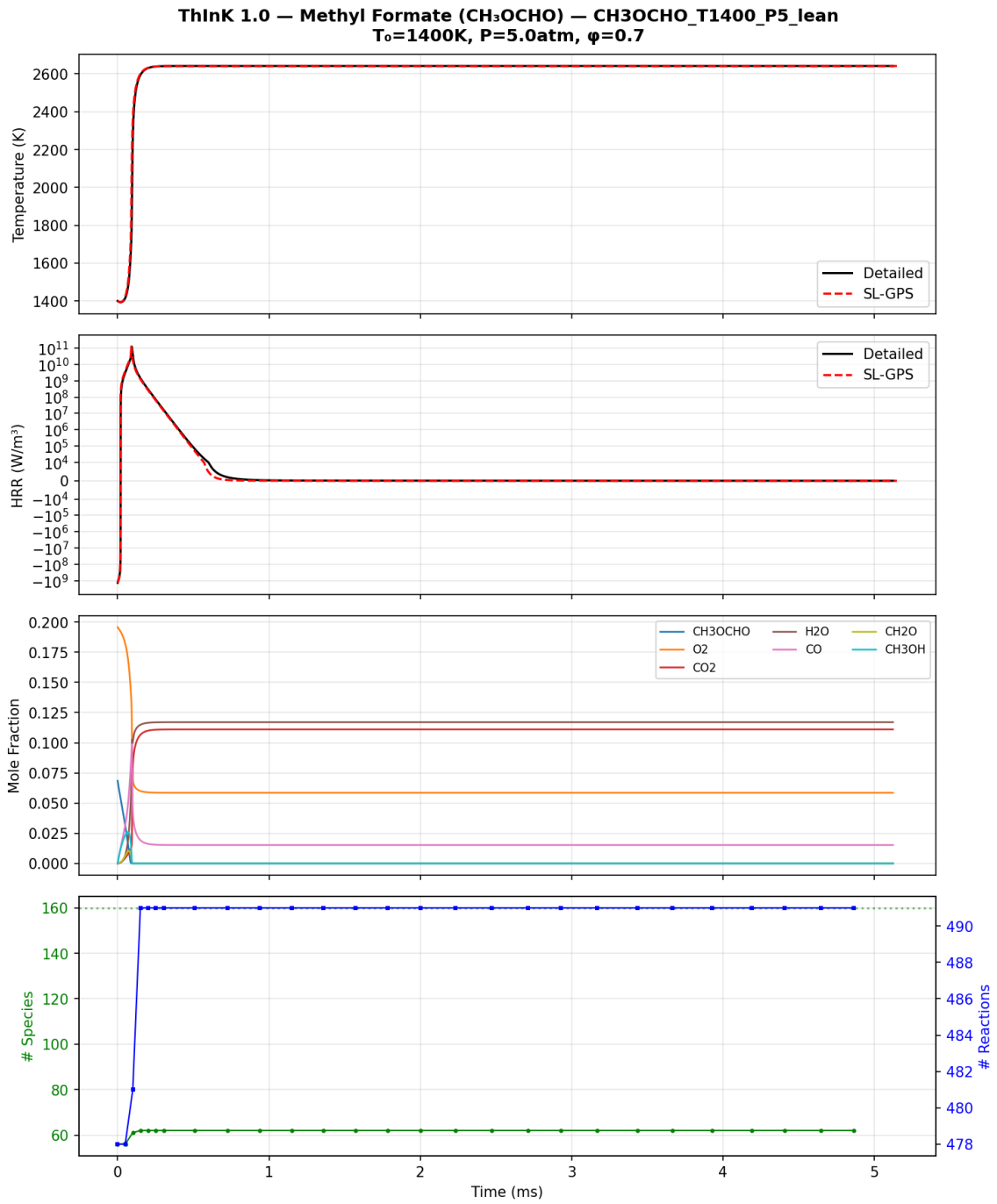


Figure 8: CH_3OCHO Case 8: $T_0 = 1400\text{ K}$, $P = 5\text{ atm}$, $\phi = 0.7$. Fuel-lean. Ignition delay error: 4.5%.

3.5 Summary

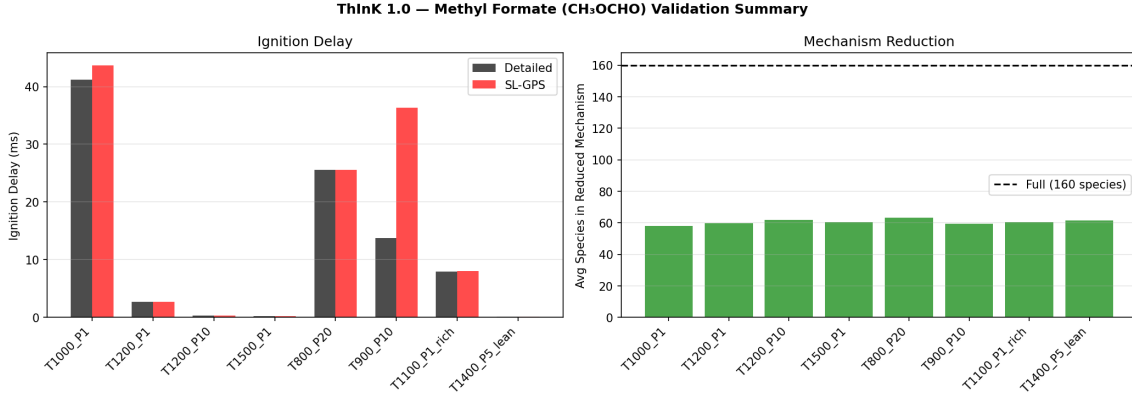


Figure 9: CH₃OCHO validation summary. Left: ignition delay comparison (detailed vs. SL-GPS). Right: average mechanism size.

Table 4: CH₃OCHO validation results summary.

Case	T (K)	P (atm)	ϕ	τ_{det} (ms)	τ_{SL} (ms)	Error (%)	Pass
T1000_P1	1000	1	1.0	41.240	43.698	6.0	✓
T1200_P1	1200	1	1.0	2.660	2.665	0.2	✓
T1200_P10	1200	10	1.0	0.280	0.290	3.4	✓
T1500_P1	1500	1	1.0	0.226	0.224	0.9	✓
T800_P20	800	20	1.0	25.529	25.597	0.3	✓
T900_P10	900	10	1.0	13.716	36.373	165.2	×
T1100_P1_rich	1100	1	1.3	7.968	8.071	1.3	✓
T1400_P5_lean	1400	5	0.7	0.094	0.090	4.5	✓
Overall				7/8 passed		22.7	
Excl. outlier (T900_P10)				7/7 passed		2.4	

Table 5: CH₃OCHO mechanism reduction summary.

Metric	Value
Detailed mechanism species	160
Mean reduced species	61
Mean reduction	62%
Always species (unconditional)	14
Variable species (ANN-predicted)	63
Never species (excluded)	83

4 Discussion

4.1 Accuracy

Across 7 of 8 validation cases, SL-GPS reproduces ignition delay times with a mean error of 2.4%, demonstrating that the ANN accurately captures the GPS species selection for methyl formate combustion. Temperature profiles, heat release rates, and major species evolution closely match the detailed mechanism.

4.2 Failed Case Analysis

The T900_P10 case ($T_0 = 900$ K, $P = 10$ atm) exhibits a 165% ignition delay error. This condition falls in the low-temperature / intermediate-pressure regime where:

- Methyl formate undergoes complex low-temperature oxidation pathways
- The competition between thermal decomposition and H-abstraction channels is sensitive to radical pool composition
- The ANN may lack sufficient training data at this specific condition

Potential remedies include:

1. Increasing `n_cases` from 15 to 30+ to improve low- T coverage
2. Adding targeted training conditions at 850–950 K, 5–15 atm
3. Widening the neural network architecture (e.g., 2×64 neurons)

4.3 Mechanism Reduction

The adaptive SL-GPS approach reduces the ThInK 1.0 mechanism from 160 species to an average of 61 active species (62% reduction). The species count varies dynamically:

- **Pre-ignition:** ~55–60 species active
- **During ignition:** peak at ~65 species as radical pool builds
- **Post-ignition:** relaxation to ~58 species

5 Reproducibility

All scripts are in `validation/ThinkMechanism/`:

```
# Full pipeline (data gen + training + validation + plots)
cd validation/ThinkMechanism
python run_pipeline_ch3ocho.py    # ~47 min total
```

5.1 Prerequisites

- Cantera $\geq 3.0.0$ with ThInK 1.0 mechanism (`ThInK_1.0_Mech.yaml`)
- Python 3.10+, TensorFlow 2.x, NumPy, scikit-learn, Matplotlib, Joblib
- ≥ 8 CPU cores recommended (parallel data generation and ensemble training)

5.2 Output Files

Table 6: Pipeline output files.

Path	Description
models/ch3ocho/model.h5	Best trained ANN model
models/ch3ocho/scaler.pkl	MinMaxScaler for input normalization
data/train_ch3ocho/data.csv	Training state vectors
data/train_ch3ocho/species.csv	GPS species importance labels
results/ch3ocho/*.png	Per-case validation plots
results/ch3ocho/*.pkl	Raw simulation data
results/ch3ocho/CH3OCHO_summary.png	Summary comparison plot
results/ch3ocho/validation_report.txt	Text report

References

- [1] ThInK 1.0: A comprehensive thermochemical and kinetic model for oxygenated C₁–C₃ fuels, *Combustion and Flame*, 2025.
- [2] R. Mishra, A. Nelson, D. Jarrahbashi, “Adaptive global pathway selection using artificial neural networks: A-priori study,” *Combustion and Flame*, vol. 244, p. 112279, 2022.
- [3] X. Gao, S. Yang, W. Sun, “A global pathway selection algorithm for the reduction of detailed chemical kinetic mechanisms,” *Combustion and Flame*, vol. 167, pp. 238–247, 2016.